Team : NEVER PAUSE UNIT

NPU Challenge

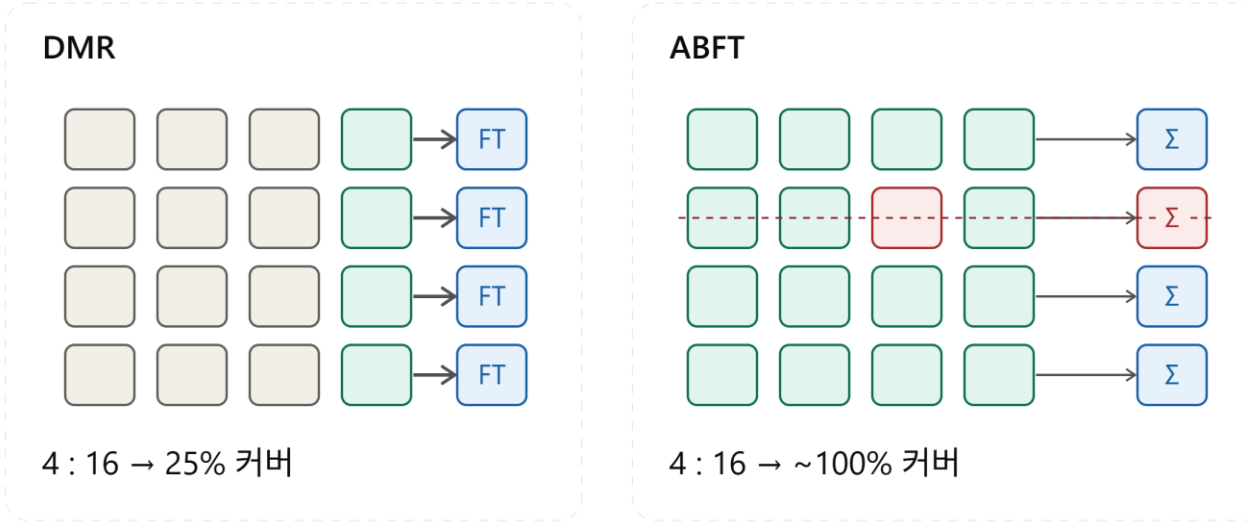
NPU : Never Pause Unit 연구 진행 발표

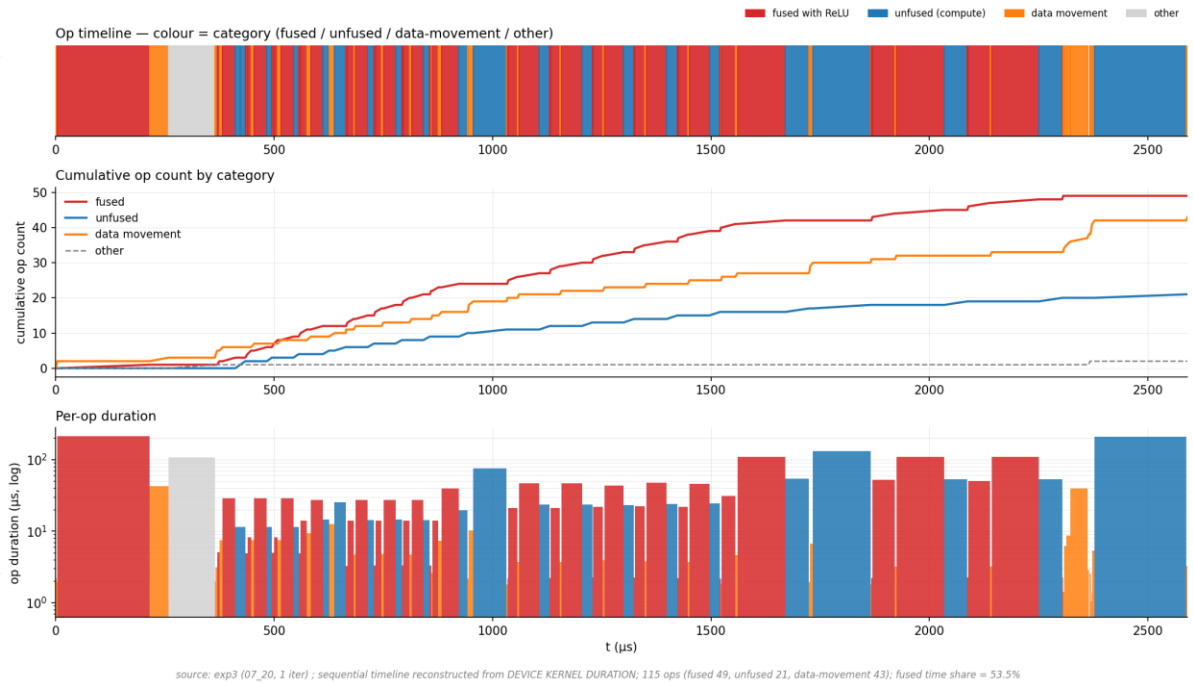
박준범 윤수민 최민기 최윤성 | 2026.07.28

Overview – New Thesis

1. Introduction & Background
2. Motivation
3. Proposed Approach
4. Evaluation Plan & Expected Results

Introduction – 기존 연구 요약





- DMR 4:16 → 25% instantaneous coverage
- ABFT : op마다 연산 구조가 선형 / 선형+비선형 / 비선형으로 달라 compute 유형별로 다른 결함 검출 알고리즘이 필요.

https://github.com/tenstorrent/tt_metal/tt-llk/docs/llk/l2/images/tensix_engine.png

Background – NBTI & HCI-induced Hardware Aging

- NBTI (Negative Bias Temperature Instability)

PMOS 트랜지스터에 음의 바이어스($V_{GS} < 0$)와 고온 스트레스가 지속적으로 인가될 때 발생

실리콘 계면($Si - SiO_2$)에 결함을 생성하여 문턱 전압(V_{th})을 상승시키는 정적 열화 메커니즘

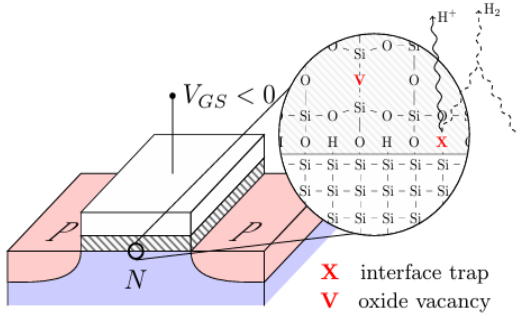
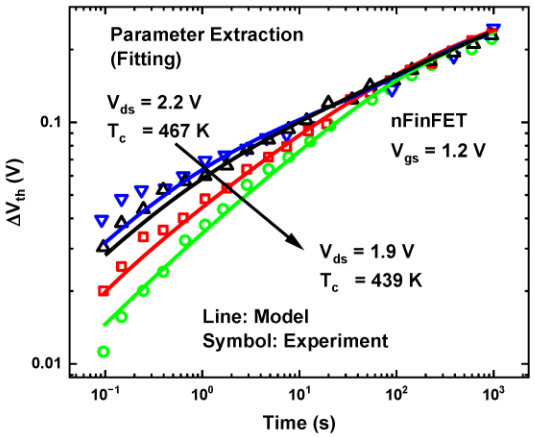


Figure 3.7.: NBTI affects PMOS transistors at the $Si - SiO_2$ interface as well as in the gate oxide

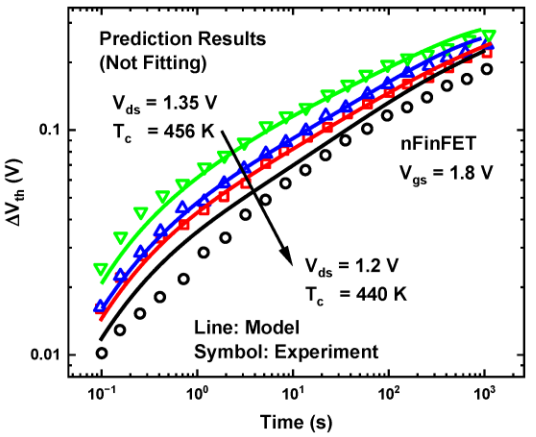
- HCI (Hot Carrier Injection)

트랜지스터의 잦은 스위칭 동작 중, 채널 내 강한 전계에 의해 고에너지 캐리어가 발생

이 캐리어들이 게이트 산화막을 뚫고 주입(Injection) 및 포획되어 소자의 특성을 훼손하는 동적 열화 메커니즘



(a)



(b)

Motivation - Aging Mitigation in Multi-core Accelerators

Aging Mitigation in Systolic Array Accelerators: Balancing PE Loads for Enhanced Reliability

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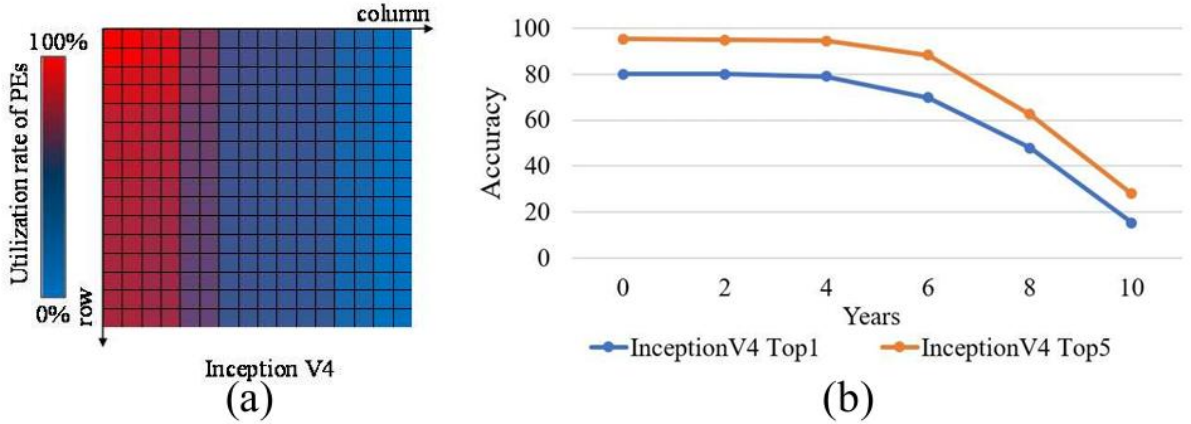


Fig. 2. (a) The utilization rate distribution of PE array with InceptionV4 (b) The accuracy of the InceptionV4 over ten years of aging.

Proactive Aging Mitigation in CGRAs through Utilization-Aware Allocation

Marcelo Brandalero*, Bernardo Neuhaus Lignati†, Antonio Carlos Schneider Beck†,
Muhammad Shafique‡, Michael Hübner*,

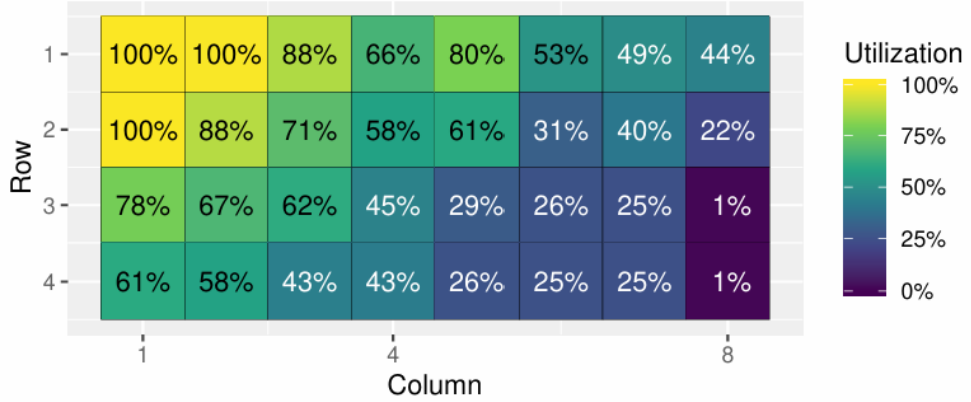


Fig. 1. Utilization rate of the FU in a 1D (only left-to-right data propagation) 4x8 CGRA design when employing traditional mapping methods and executing a set of embedded applications.

멀티코어/배열 아키텍처에서의 Utilization Imbalances는 시스템의 장기 신뢰성을 파괴

Motivation - Aging Mitigation in Multi-core Accelerators

[13] Henkel, Jörg, et al. "Thermal management for dependable on-chip systems." 2013 18th Asia and South Pacific Design Automation Conference (ASP-DAC). IEEE, 2013.

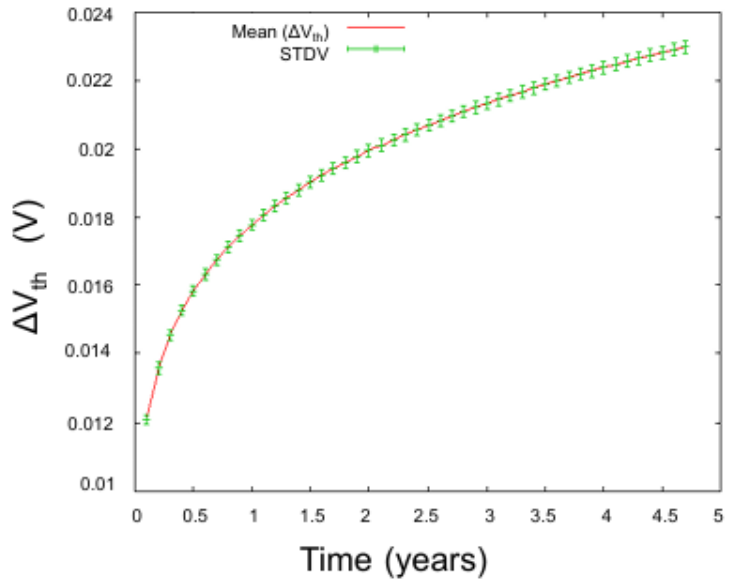


Figure 3.1.: NBTI induced shift in threshold voltage over time in 65nm latch

$$\Delta V_{th,NBTI} = A_{NBTI} \times e^{(\frac{-E_G}{kT})} \times V_{dd}^4 \times t^{1/6} \times u^{1/6}$$

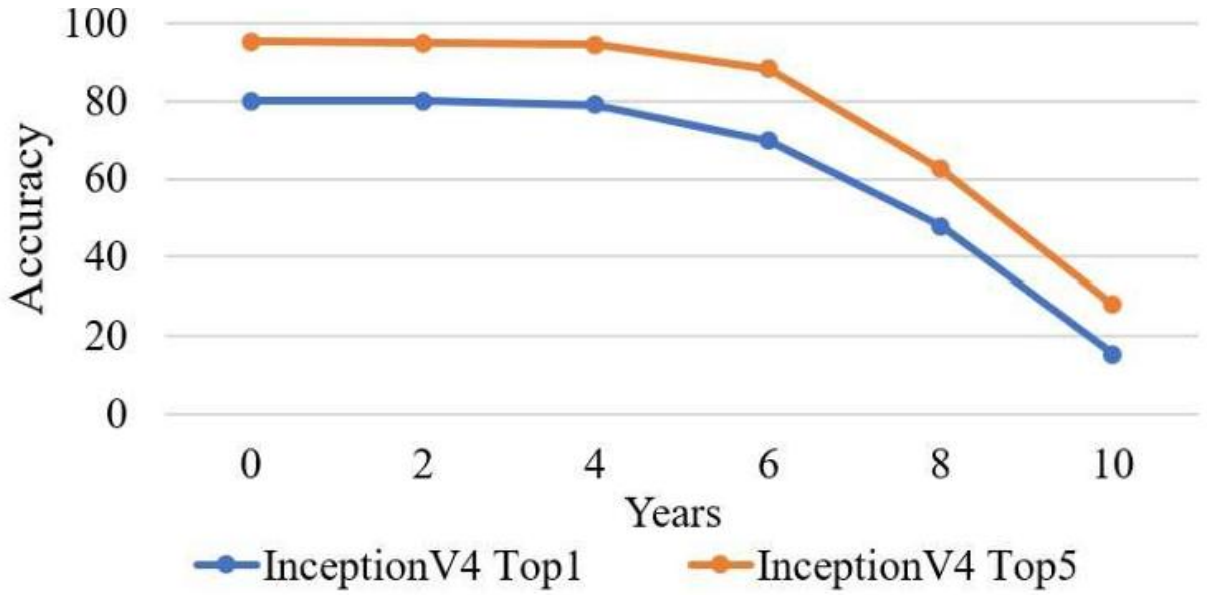
$$\Delta V_{th,HCI} = A_{HCI} \times u \times f \times e^{(\frac{-E_b}{kT})} \times t^{1/6}$$

Aging Mitigation in Systolic Array Accelerators: Balancing PE Loads for Enhanced Reliability

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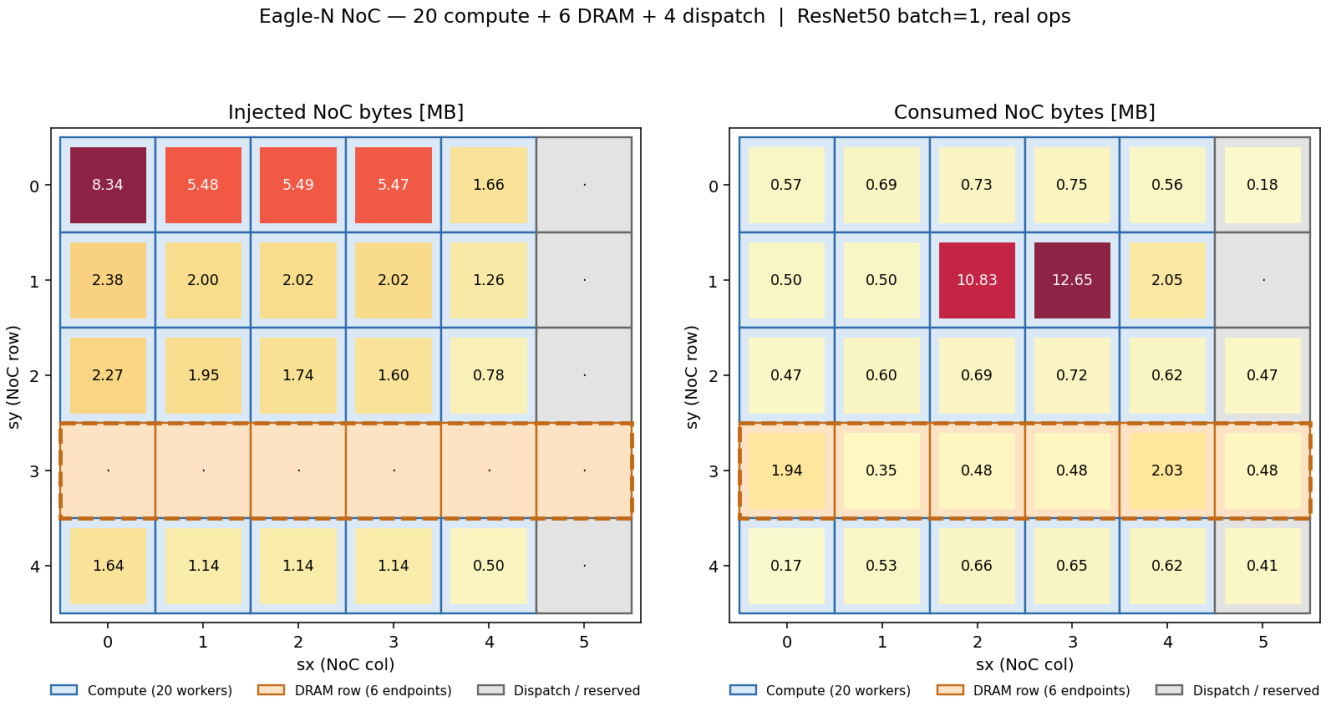
Jing-Yang Jou
Department of Electrical Engineering
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Taoyuan, Taiwan
jyjou@ee.ncu.edu.tw

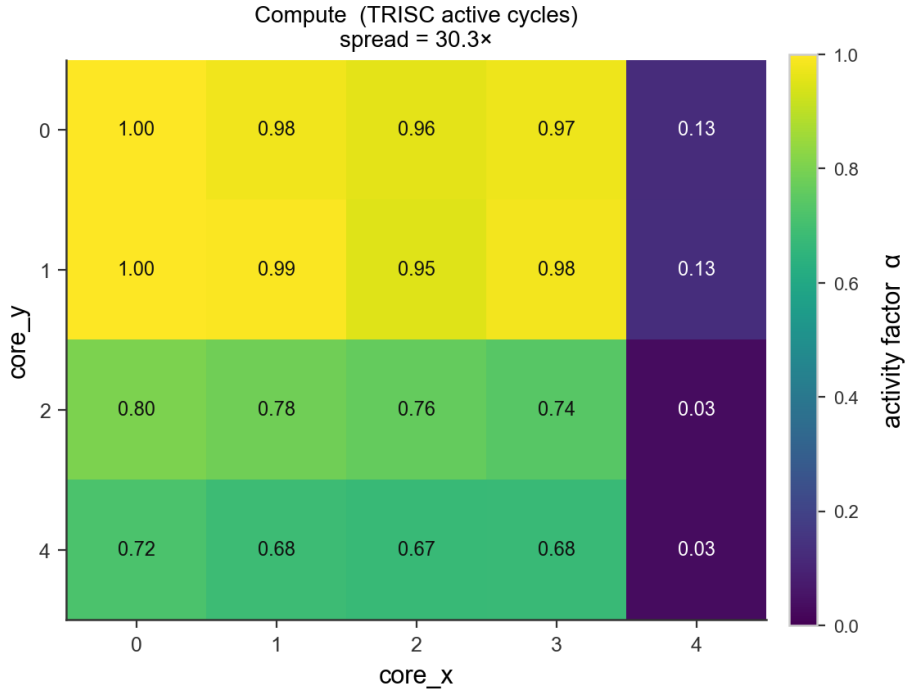


하드웨어 시뮬레이터 기반 Accuracy 추정

Motivation - Aging Mitigation in Multi-core Accelerators

Research Gap : Scalability of Aging-Aware Workload in NoC-based Programmable NPUs.





Physical layout: y=0,1,2 compute stacked → y=3 DRAM row → y=4 compute. Rows y=2 and y=4 are 2 NoC hops apart (must cross DRAM row).

Motivation - Aging Mitigation in Multi-core Accelerators

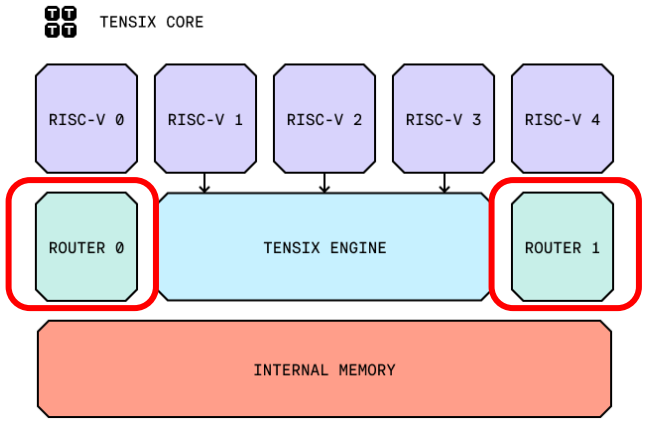
Research Gap : Scalability of Aging-Aware Workload in NoC-based Programmable NPUs.

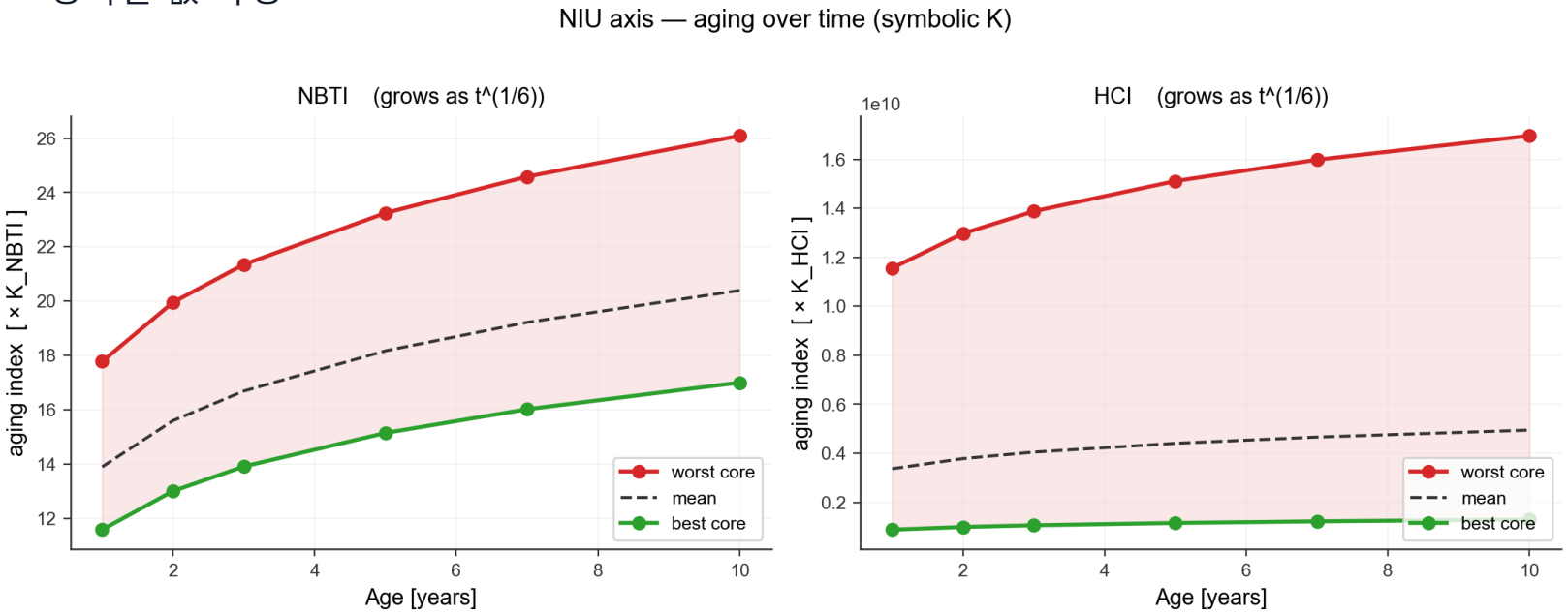
$$\Delta V_{th,NBTI} = A_{NBTI} \times e^{\left(\frac{-k_G}{kT}\right)} \times V_{dd}^4 \times t^{1/6} \times u^{1/6}$$

* 정확한 값 측정 X

$$\Delta V_{th,HCI} = A_{HCI} \times u \times f \times e^{\left(\frac{-E_b}{kT}\right)} \times t^{1/6}$$

* 정확한 값 측정

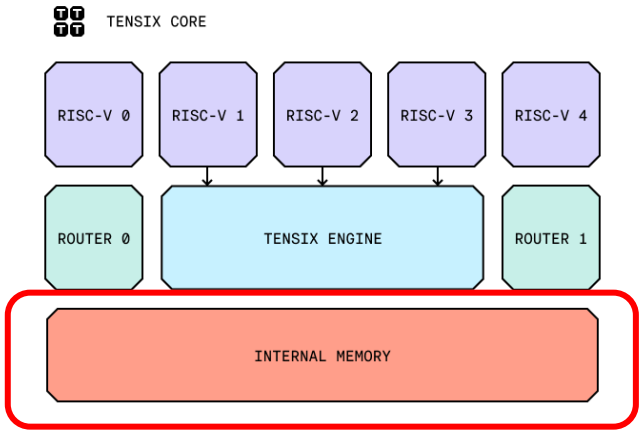


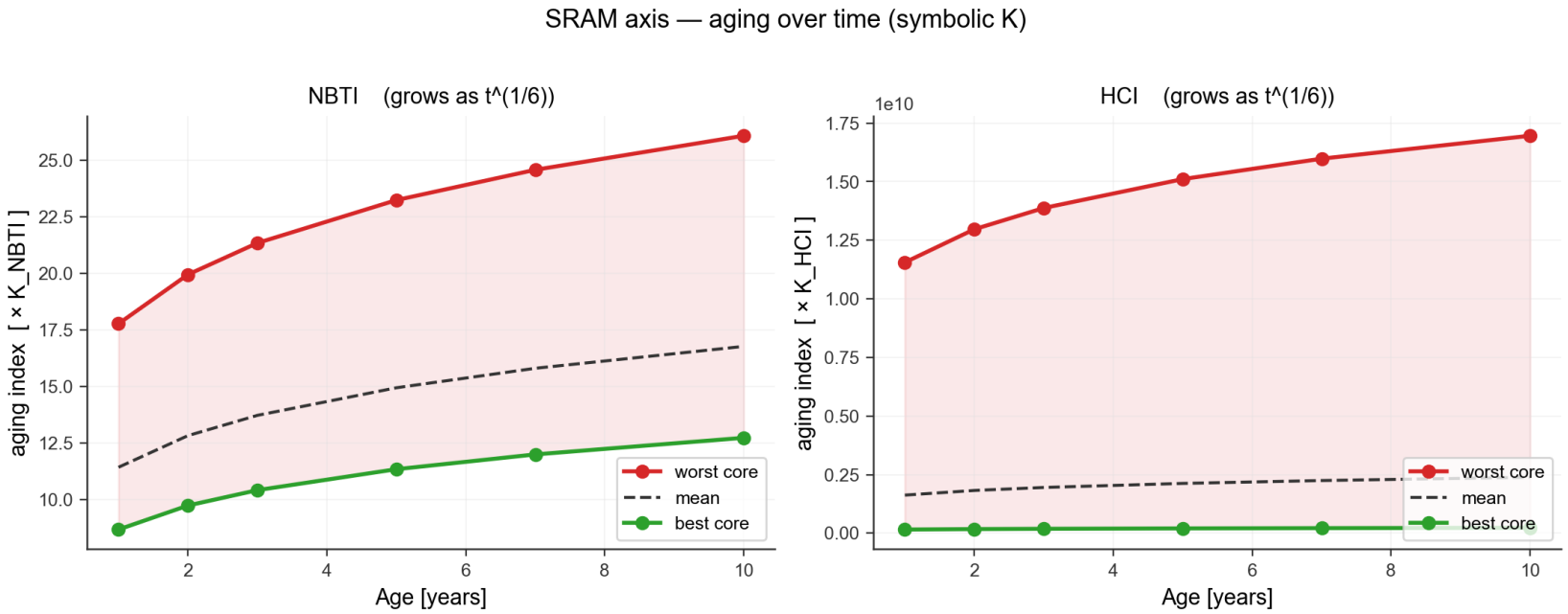


Motivation - Aging Mitigation in Multi-core Accelerators

$$\Delta V_{th,NBTI} = A_{NBTI} \times e^{\left(\frac{-k_G}{kT}\right)} \times V_{dd}^4 \times t^{1/6} \times u^{1/6}$$

$$\Delta V_{th,HCI} = A_{HCI} \times u \times f \times e^{\left(\frac{-E_b}{kT}\right)} \times t^{1/6}$$

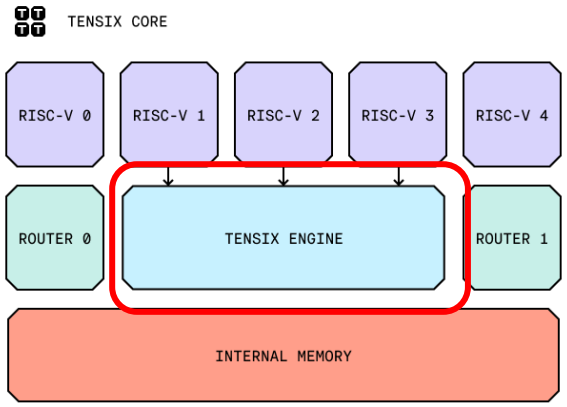


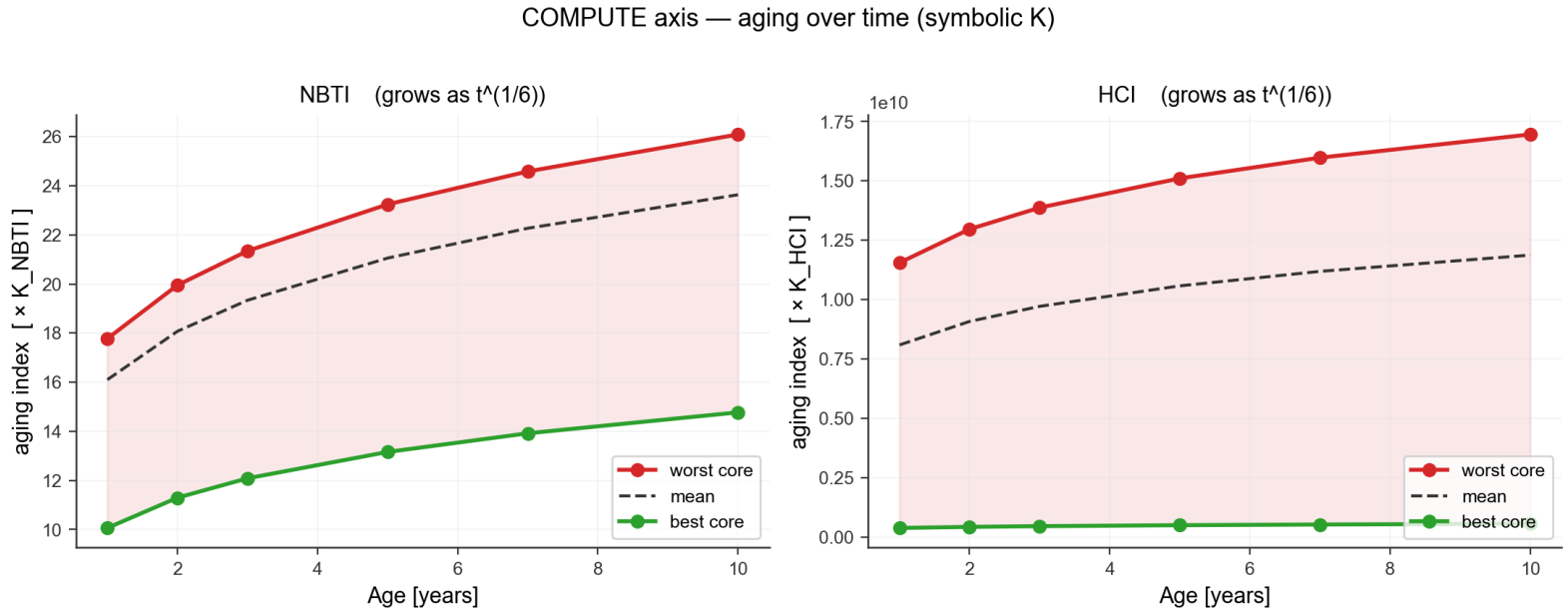


Motivation - Aging Mitigation in Multi-core Accelerators

$$\Delta V_{th,NBTI} = A_{NBTI} \times e^{(\frac{-k_G}{kT})} \times V_{dd}^4 \times t^{1/6} \times u^{1/6}$$

$$\Delta V_{th,HCI} = A_{HCI} \times u \times f \times e^{(\frac{-E_b}{kT})} \times t^{1/6}$$





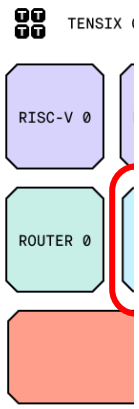
Motivation - Aging Mitigation in Multi-core Accelerators

$$\Delta V_{th,NBTI} = A_{NBTI} \times e^{(\frac{-k_G}{kT})} \times V_{dd}^4 \times t^{1/6} \times u^{1/6}$$

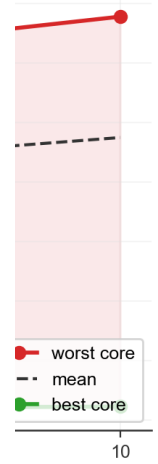
$$\Delta V_{th,HCI} = A_{HCI} \times u \times f \times e^{(\frac{-E_b}{kT})} \times t^{1/6}$$

Worst / best aging ratio per axis (baseline, ResNet50 batch=1)

NBTI compresses activity imbalance by 6th-root; HCI carries it linearly → HCI reveals the true aging skew.

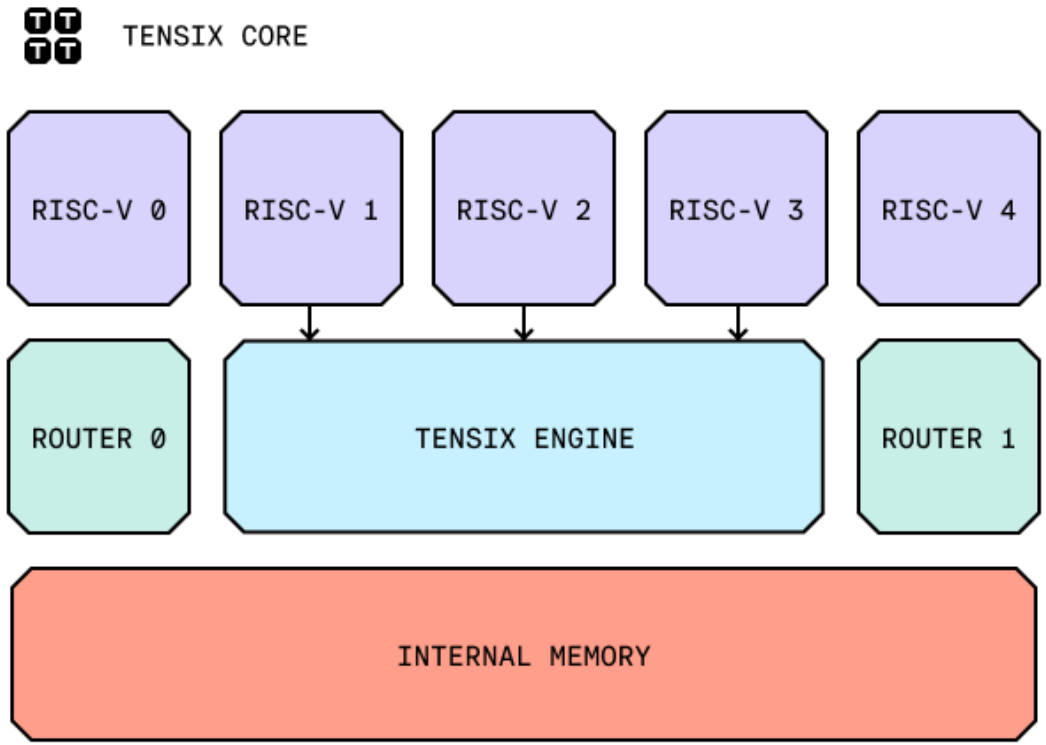


Axis	Raw activity (worst / best)	NBTI aging ratio = raw^(1/6)	HCI aging ratio = raw
NIU	13.08×	1.54×	**13.08×**
SRAM	74.14×	2.05×	**74.14×**
Compute	30.35×	1.77×	**30.35×**



Read: SRAM worst-core wears 74× faster than SRAM best-core (HCI-dominant regime). System lifetime = worst-core lifetime.

Motivation - Aging Mitigation in Multi-core Accelerators



1. Component Stress Analysis

Tensix 코어 내 NIU·SRAM·Compute의 컴포넌트별 스트레스 분석

2. Component-Aware Aging Simulation

컴포넌트별 열화 특성을 반영한 정밀 장기 노화 시뮬레이션

3. Lifetime & Accuracy-Aware Mapping

하드웨어 마모와 AI 정확도 저하를 방어하는 최적의 워크로드 매핑)

Proposed Approach - Asymmetric Stress

```yaml
grid:
  x_size: 6
  y_size: 5

arc:
  [ ]

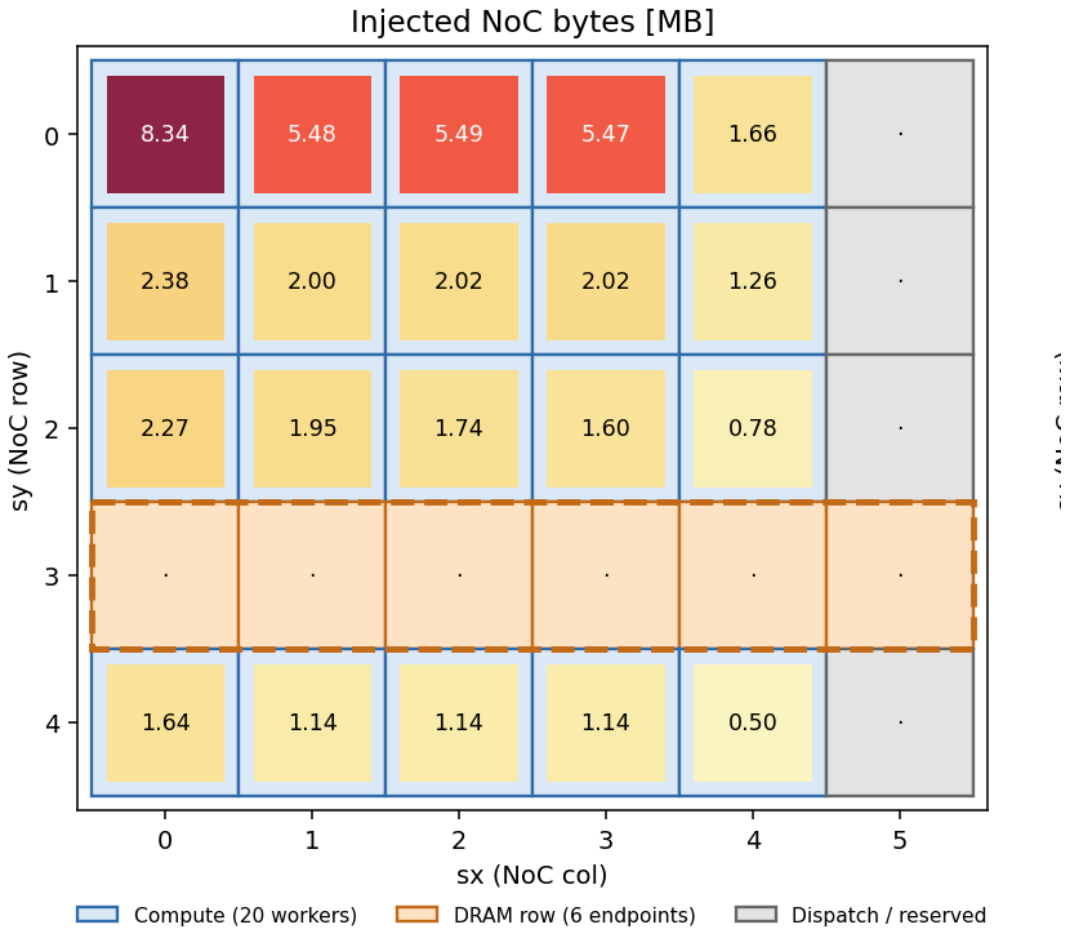
pcie:
  [ ]

dram:
  [
      [ 0-3, 1-3 ], #,,
      [ 4-3, 2-3 ],
      [ 5-3, 3-3 ]
eth:
  []

functional_workers:
  [
   0-0,  1-0,  2-0, 3-0, 4-0, 5-0,
   0-1,  1-1,  2-1, 3-1, 4-1, 5-1,
   0-2,  1-2,  2-2, 3-2, 4-2, 5-2,
   0-4,  1-4,  2-4, 3-4, 4-4, 5-4,
  ]

harvested_workers:
  []

router_only:
  []
```



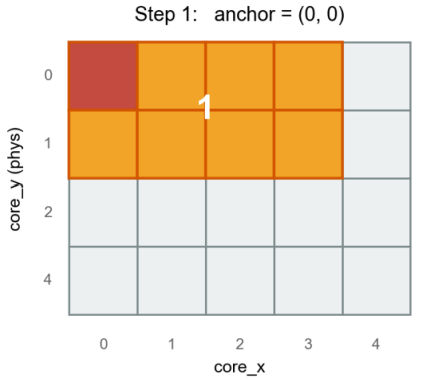
Pysical Constraint : Row 3에 위치한 dram이 Compute core의 비대칭을 만들어냄.

Proposed Approach - Baseline Mapping & Routing Logic

TT-metal Logic : Anchor 하드코딩 conv/conv2d/device/conv2d_op_sharded_program_factory.cpp

```
576     const CoreCoord top_left_core = {(std::size_t)0, (std::size_t)0};
577     const CoreCoord top_left_core_plus_one = {(std::size_t)1, (std::size_t)1};
578     const CoreCoord bottom_right_core = {(std::size_t)in_num_cores_x - 1, (std::size_t)in_num_cores_y - 1};
579     const CoreCoord top_left_core_physical = device->worker_core_from_logical_core(top_left_core);
580     const CoreCoord top_left_core_plus_one_physical = device->worker_core_from_logical_core(top_left_core_plus_one);
581     const CoreCoord bottom_right_core_physical = device->worker_core_from_logical_core(bottom_right_core);
```

```
239     CoreRangeSet all_cores;
240     if (height_sharded) {
241         all_cores = CoreRangeSet(CoreRange(
242             CoreCoord(0, 0),
243             CoreCoord(parallelization_config.grid_size.x - 1, parallelization_config.grid_size.y - 1)));
244     } else {
245         all_cores = input_cores.merge(output_cores);
246     }
```



시작 좌표(Anchor) 하드코딩의 한계

경직된 사각형(Rectangle) 할당 제약

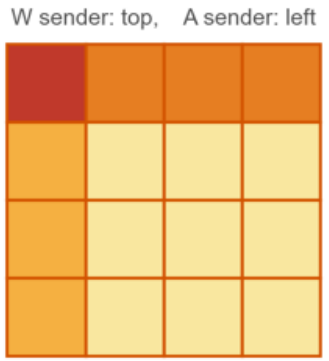
동적 워크로드 이동(Shift) 불가

(0,0) 코어 주변에 극심한 노화(Aging Hotspot) 집중

Proposed Approach - Baseline Mapping & Routing Logic

TT-metal Logic : Mcast sender/receiver 사각형 강제
/conv/conv2d/device/conv2d_op_sharded_program_factory.cpp

```
596     if (block_sharded) {
597         const CoreCoord out_bottom_right_core = {(std::size_t)num_cores_x - 1, (std::size_t)num_cores_y - 1};
598         // 2D mcast
599         if (transpose_mcast) {
600             mcast_sender_cores = CoreRangeSet(CoreRange(top_left_core, CoreCoord(0, num_cores_y - 1)));
601             if (!skip_weights_mcast) {
602                 mcast_receiver_cores = CoreRange(CoreCoord(1, 0), out_bottom_right_core);
603             }
604         } else {
605             mcast_sender_cores = CoreRangeSet(CoreRange(top_left_core, CoreCoord(num_cores_x - 1, 0)));
606             if (!skip_weights_mcast) {
607                 mcast_receiver_cores = CoreRange(CoreCoord(0, 1), out_bottom_right_core);
608             }
609         }
610         if (populate_skipped_work_cores) {
611             mcast_sender_cores = input_cores.subtract(mcast_receiver_cores);
612         }
```



Sender 좌표 하드코딩
라우팅 트래픽 교차 병목
Receiver 사각형 강제

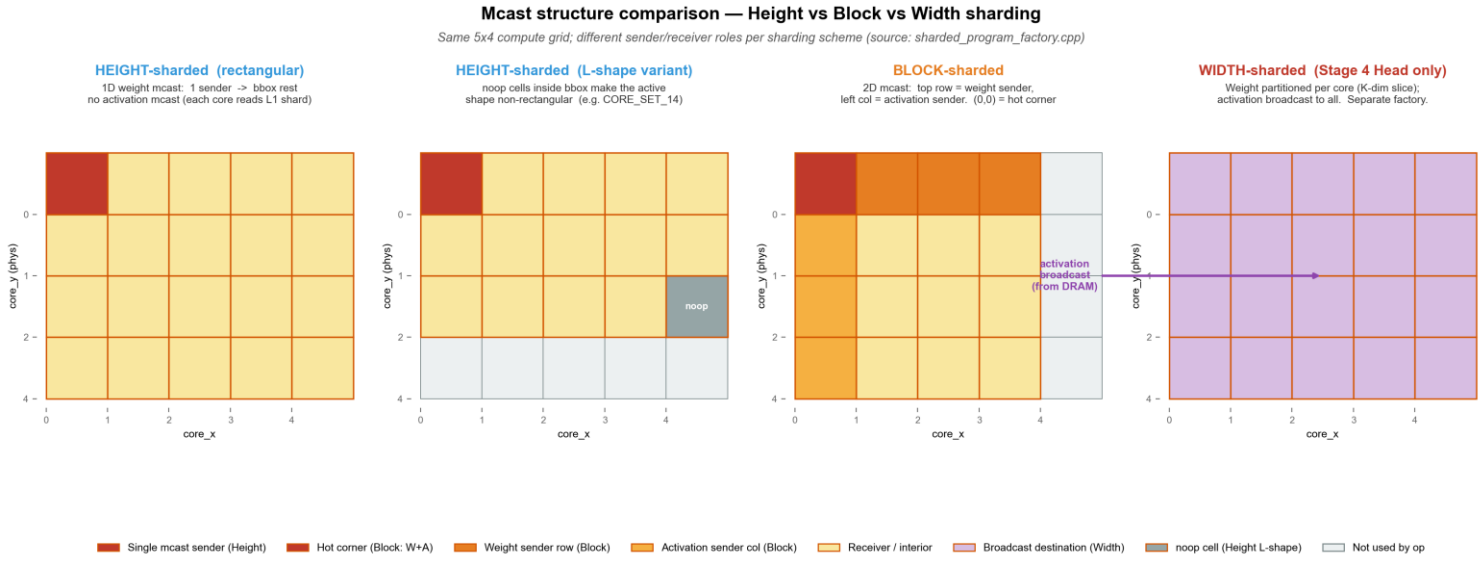
(0,0) 및 엣지 코어의 극단적인 통신 열화 가속

Proposed Approach - Baseline Mapping & Routing Logic

TT-metal Logic : Mcast conv/conv2d/device/conv2d_op_sharded_program_factory.cpp

Sharding scheme comparison

	Height	Block	Width
Mcast pattern	1D weight (single sender)	2D (weight + activation)	broadcast only
Sender location	top-left core	top row + left col	(no fan-out sender)
Rotation freedom	anchor shift + L-shape	anchor shift + 4-reflection	N/A
In rotation scope?	YES	YES	NO

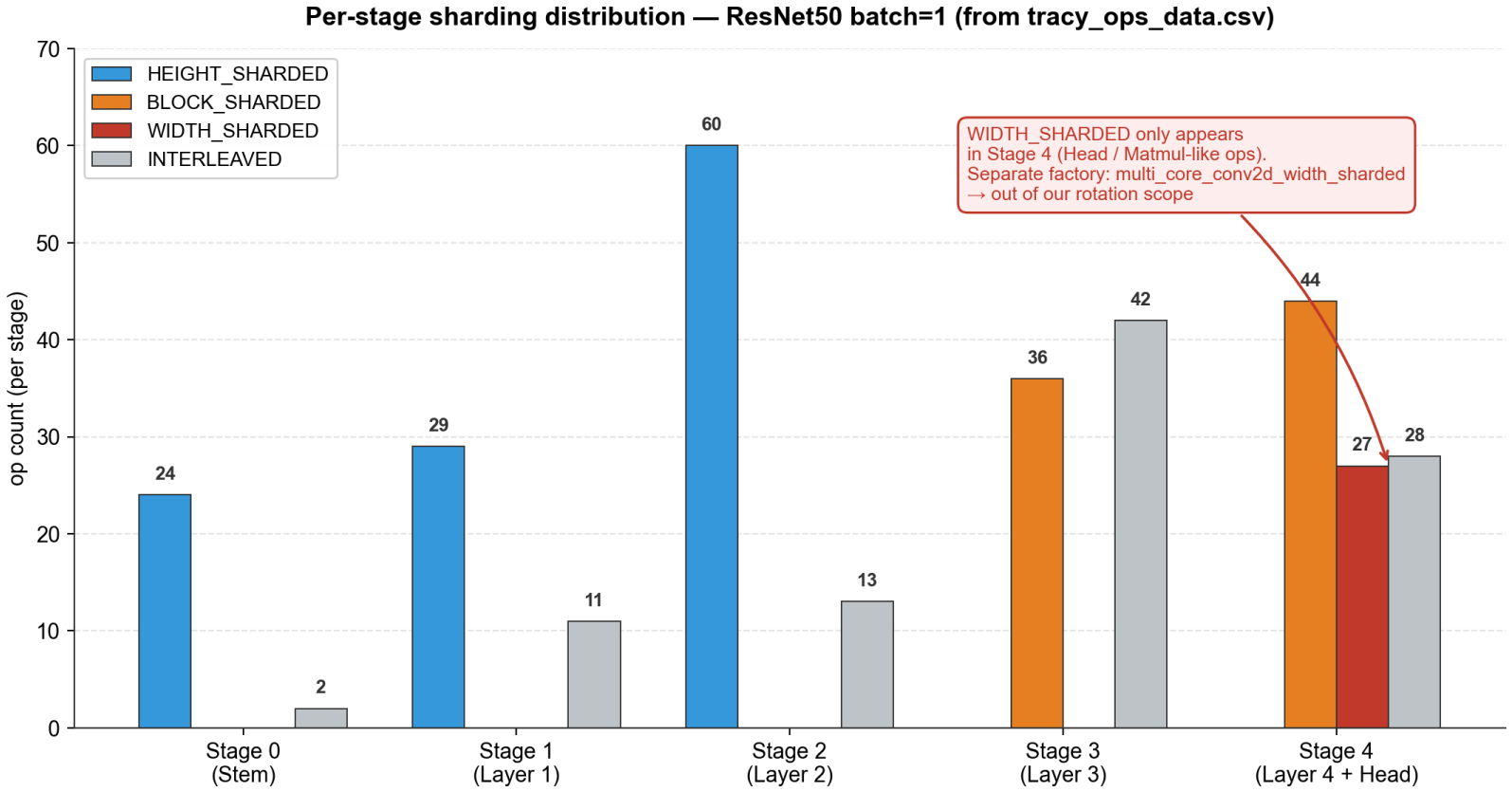


Evaluation Plan & Expected Results

Rotationing Examples

Evaluation Plan & Expected Results

Workload Dependency : Sharding Strategy Analysis on ResNet50



HEIGHT (Layer 1-2) and BLOCK (Layer 3-4) share the same sharded_program_factory.cpp — our rotation targets. INTERLEAVED = DRAM tensor layout (not compute sharding).

매핑 제어의 복잡성 증가 : 샤딩 기법에 따fk 코어를 배치하고 라우팅하는 형태가 완전히 달라지므로 단순한 단일 매핑 규칙만으로는 전체 칩의 노화를 제어할 수 없음.

Tier 1 rotation: cyclic anchor rotation

Each successive op executes at the next anchor in cycle. Shape preserved, only pivot shifts.

Op shape: 4 x 2



Valid anchors on 5x4 grid:

x in 0..1, y in 0..2

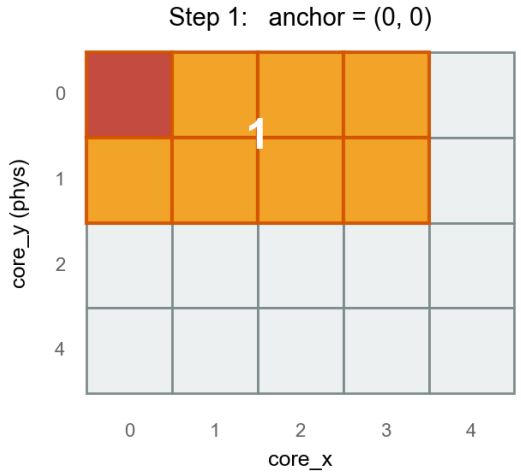
= 2 x 3 = 6 positions

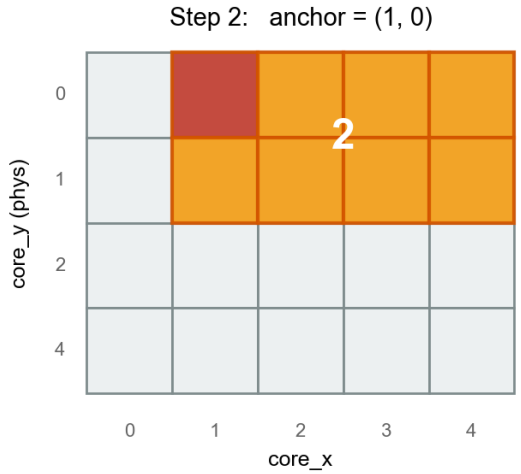
Tier 1 policy

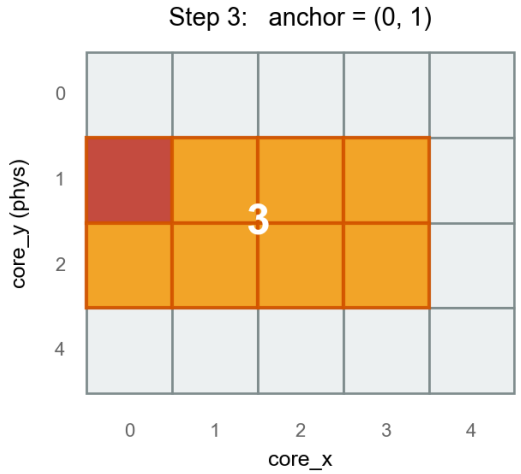
Cyclic anchor rotation:

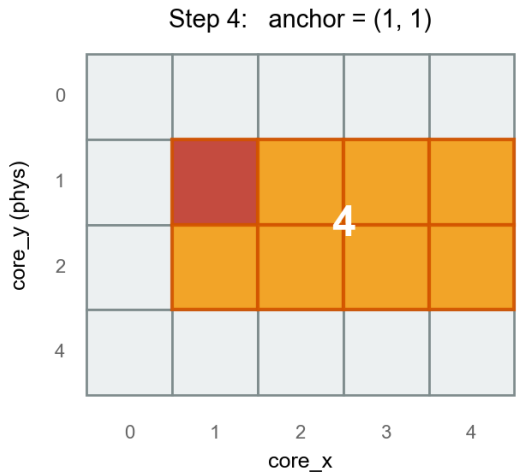
sweep the 6 anchors in order.

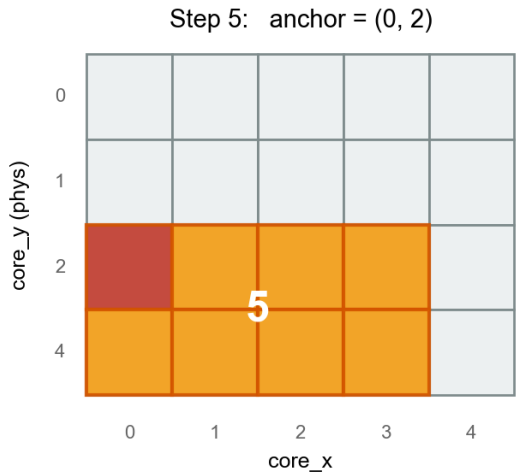
Op #k -> anchor[k mod 6]

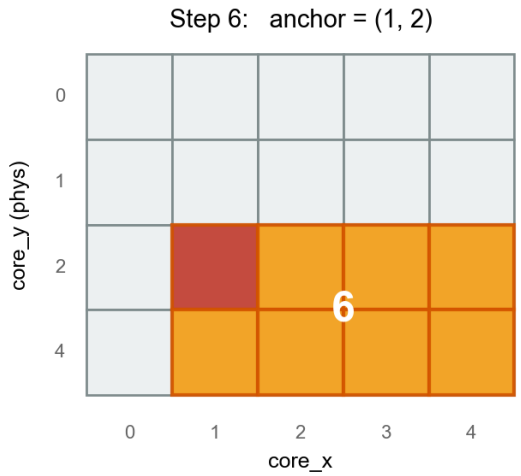










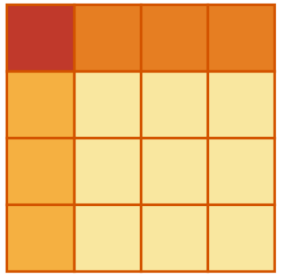


Tier 2 rotation: anchor shift + reflection (block-sharded)

4x4 shape on 5x4 grid: 2 anchors x 4 reflections = 8 configs. Reflection swaps which edge is the mcast sender (hot corner moves).

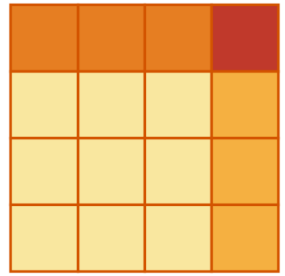
TL (default)

W sender: top, A sender: left



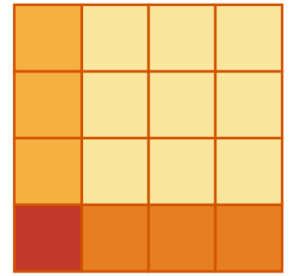
TR (flip H)

W sender: top, A sender: right



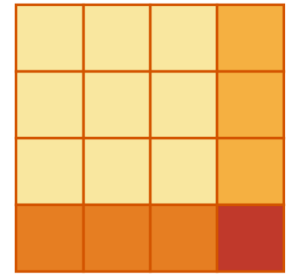
BL (flip V)

W sender: bottom, A sender: left



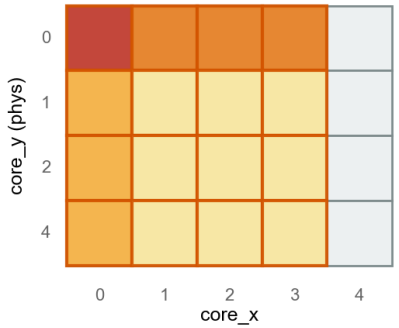
BR (flip both)

W sender: bottom, A sender: right

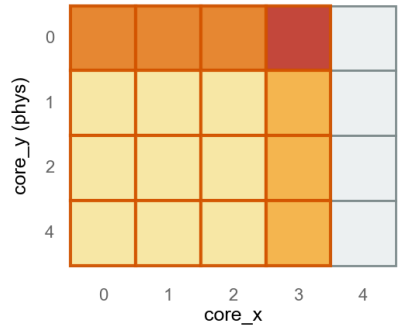


v v v Apply each reflection at 2 valid anchors v v v

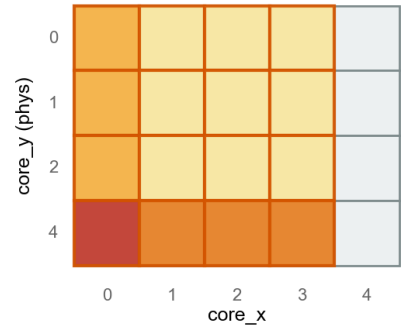
Step 1: anchor=(0,0) + TL



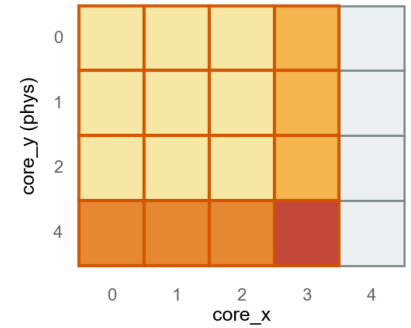
Step 2: anchor=(0,0) + TR



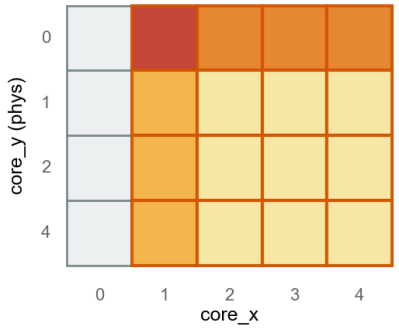
Step 3: anchor=(0,0) + BL



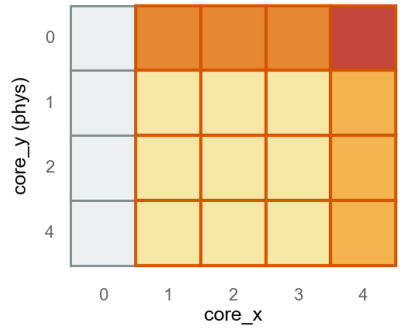
Step 4: anchor=(0,0) + BR



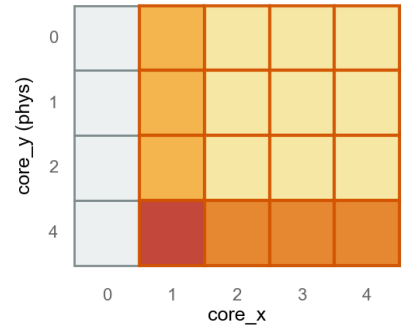
Step 5: anchor=(1,0) + TL



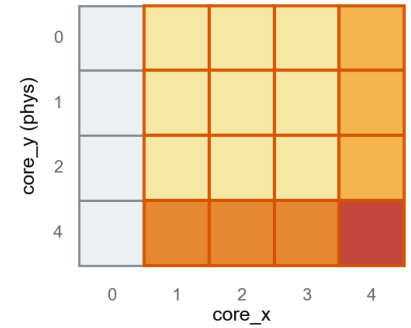
Step 6: anchor=(1,0) + TR



Step 7: anchor=(1,0) + BL



Step 8: anchor=(1,0) + BR



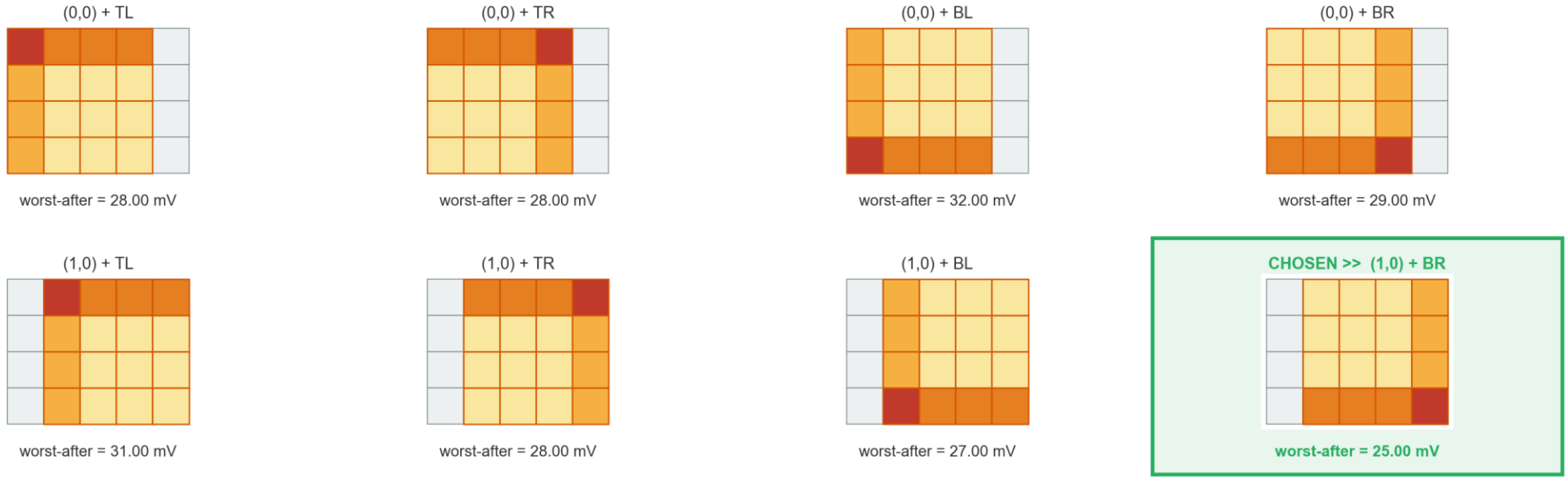
Hot corner (W+A sender intersection) | Weight mcast sender (row) | Activation mcast sender (col) | Interior (receiver only)

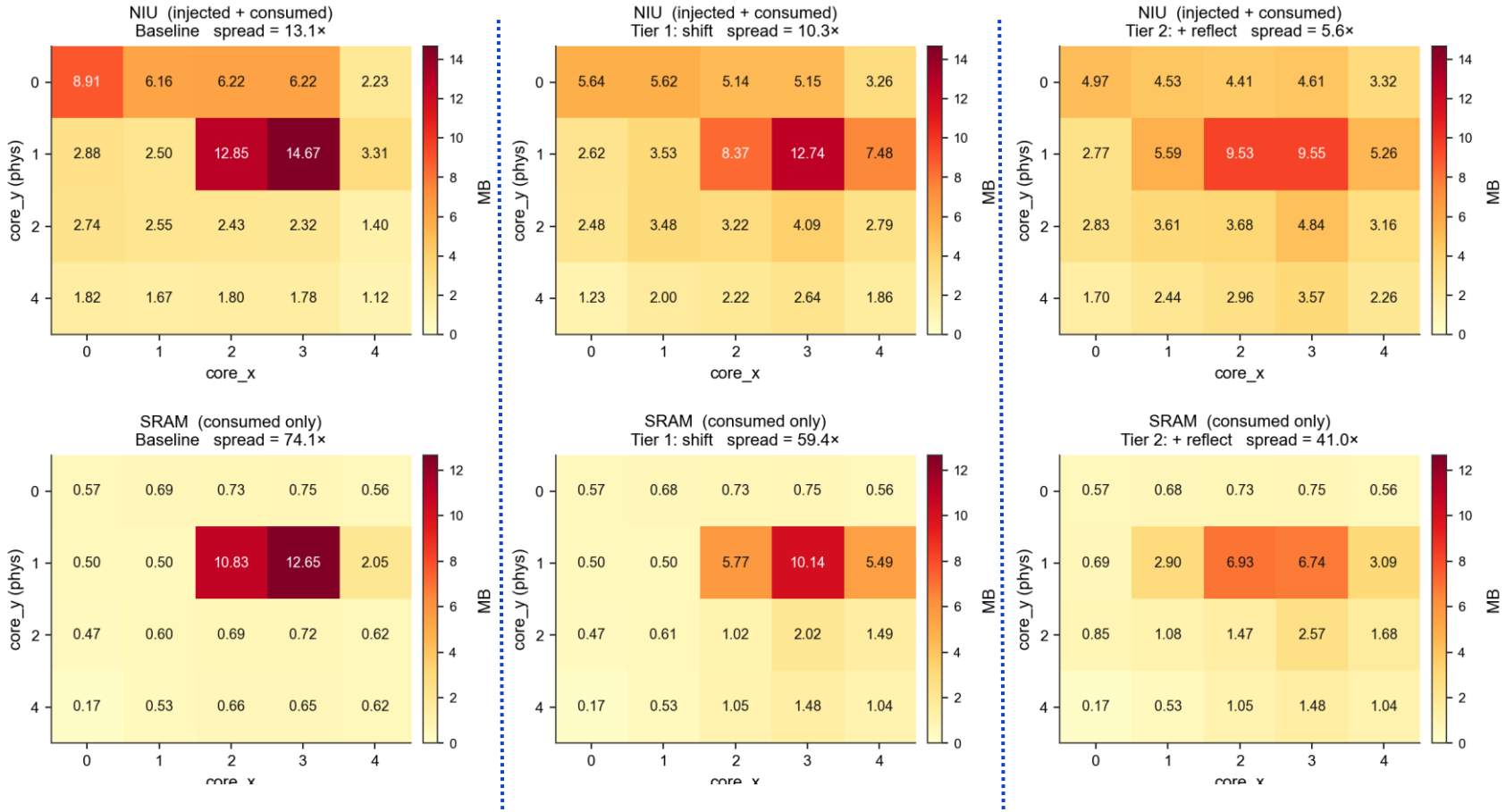
Tier 3 rotation: aging-aware greedy (min-worst config selection)

For each op: enumerate all valid configs, predict per-core Vth after placement, pick argmin(worst-core).



(b) 8 candidates and their scores — green box = argmin





Rotation tier summary — worst-core & spread reduction vs baseline

Source: rotation_tier{1,2,3}_per_core.csv (NIU = total NoC bytes, SRAM = L1 consumed bytes)

	NIU worst-core	NIU spread	SRAM worst-core	SRAM spread
Baseline	14.67 MB	13.08×	12.65 MB	74.14×
Tier 1	12.74 MB (−13%)	10.32× (−21%)	10.14 MB (−20%)	59.43× (−20%)
Tier 2	9.55 MB (−35%)	5.61× (−57%)	6.93 MB (−45%)	41.01× (−45%)
Tier 3	**6.76 MB** (−54%)	**5.25×** (−60%)	**4.82 MB** (−62%)	**28.05×** (−62%)

Next Week Plan

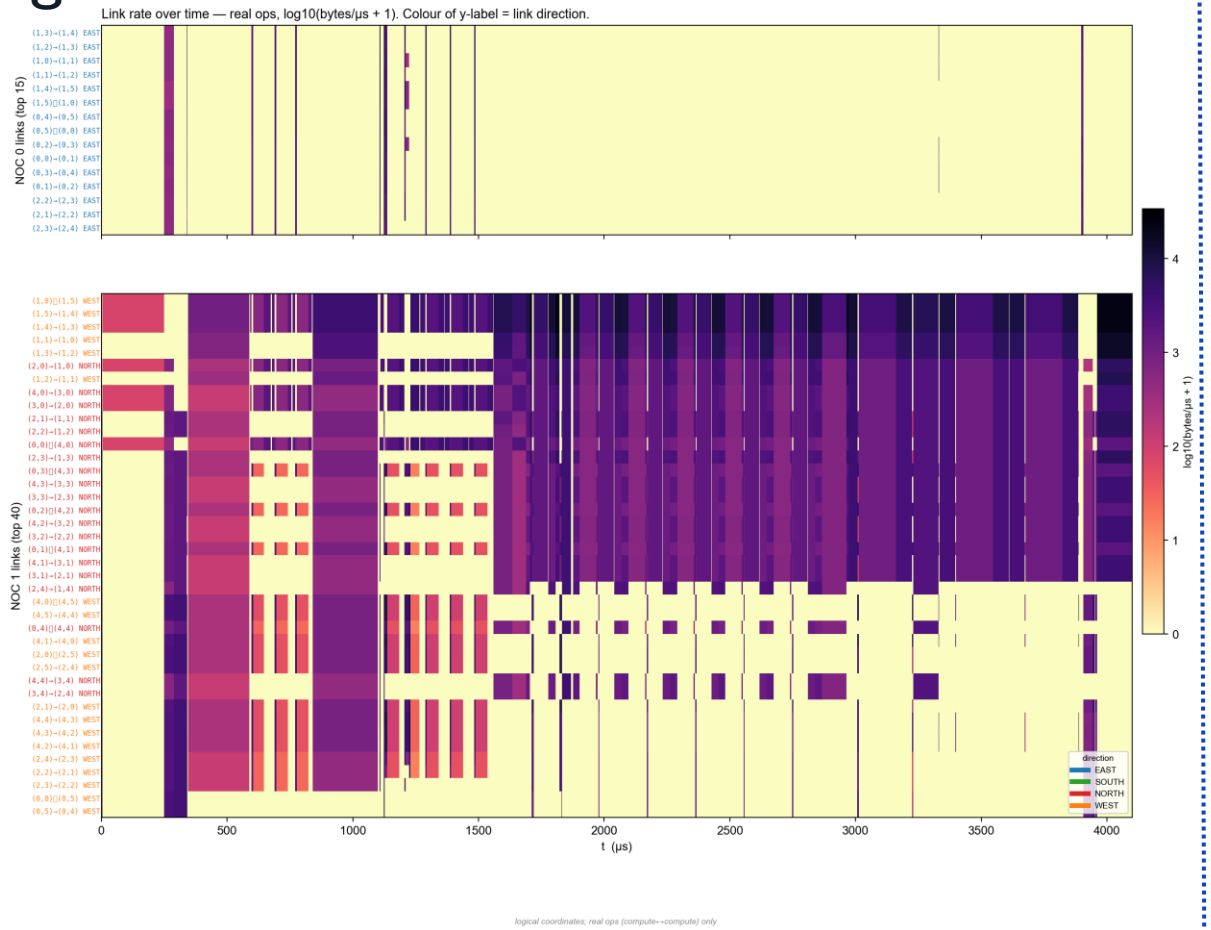
[추가 분석 대상] NoC0/1 채널 간 트래픽 비대칭성

NoC0

(Read)

NoC1

(Write)



- Aging 시뮬레이터 연동 및 장기 스트레스에 따른 AI 모델 Accuracy 저하 검증
- TT-Metalium 코어 매핑 API 수정
- 워크로드 Sharding 특성을 고려한 코어 Rotaion 알고리즘 설계

향후 계획

~~1주차 · HW 아키텍쳐 학습~~

~~2주차 · SW 스택 이해~~

~~3주차 · AI model 포로파일링 및 연구주제 탐구~~

~~4주차 · 여유 코어 확보 및 SubDevice 탐구~~

5주차 ·
여유 코어에 서브디바이스 적용(Fault detect logic)
- 연산 결과에 치명적인 요소들 탐색

코어 제한 모델 최적화
- 베이스라인 모델 수준으로 최적화